Paper 7 — Flatland: Large-Step GD

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Flatland: The Adventures of Gradient Descent with Large Step Sizes

Papers Master: `galli2026flatland` **Tags:** D, G, E / R, B | **Must read (main body)**

The paper in one minute

Question. How can GD use extremely large learning rates yet converge, and must the flattest neural-network regions be desirable?
Model / Method. First-order adaptive step sizes via nonmonotone *equality line search*; theory uses local smoothness along the gradient direction.
Finding. Large steps often put training at the edge of stability (EoS), with oscillating loss. In the MSE-classification experiments, some runs reach maximally flat *saddle* regions early and fail to learn; limiting step size improves them.
Why we care. Learning-rate choice, curvature, and trainability are inseparable. Neither monotone loss decrease nor maximal flatness is always a useful training objective.

1. Start with the mystery: large steps do not always fail

Traditional GD uses $w_{k+1} = w_k - \eta_k \nabla f(w_k)$. A globally L-Lipschitz gradient suggests using a small step $\eta < 2/L$ to secure descent. But a deep network need not have one finite global L, and practitioners often train with larger steps. This paper asks how to choose such steps without requiring global smoothness. **A worked example: why the edge threshold is 2**

Take the *teaching example* $f(x) = \frac{1}{2}\lambda x^2$ with $\lambda = 4$, so $f'(x) = 4x$. GD becomes

$$x_{k+1} = x_k - \eta\lambda x_k = (1 - 4\eta)x_k.$$

To contract toward $x = 0$, we need $|1 - 4\eta| < 1$, i.e. $0 < \eta\lambda < 2$. For $\eta = 0.40$, the multiplier is -0.6: each step crosses the optimum, but gets closer. At $\eta = 0.50$, it is -1: it oscillates forever. For $\eta = 0.55$, the multiplier is -1.2, so *fixed-step GD on this quadratic diverges*.

Read this carefully. The empirical EoS diagnostic is approximately $\eta_k \lambda_{\max}(\nabla^2 f(w_k)) \simeq 2$, with possible excursions above 2 in nonquadratic training. This is *not* a claim that holding $\eta\lambda > 2$ fixed is safe on every objective.

What "sharpness" means here

$\lambda_{\max}(H(w))$ is the largest eigenvalue of $H(w) = \nabla^2 f(w)$. It measures the strongest *positive* local curvature. A low sharpness value does not by itself imply a low loss, a local minimum, or good test accuracy.

2. What the authors actually change: the line search

Instead of knowing a global Lipschitz constant, the paper measures how much the gradient changes *along a proposed GD segment*:

$$\ell(w,\eta) = \sup_{0<t\le\eta} \frac{\|\nabla f(w) - \nabla f(w - t\nabla f(w))\|}{\|t\nabla f(w)\|}.$$

This **local segment smoothness** need only be finite locally. The paper also analyzes Hölder versions with gradient differences of order $\|u-v\|^\nu$, $0 \le \nu \le 1$. **A precise definition of a "large" step**

For a fixed window length P, a sequence is large if each $\eta_k \ge 1/\ell(w_k, \eta_{w_k})$ and at least once within the next P steps, $\eta_{k+p} > 2/\ell(w_{k+p}, \eta_{w_{k+p}})$. The point is *persistent aggressive stepping*, not merely a large numerical learning rate. **Nonmonotone equality line search (NLS)**

The acceptance test is

$$f(w_k - \eta_k g_k) \le R_k - c\eta_k \|g_k\|^2, \quad g_k = \nabla f(w_k), \quad R_k \ge f(w_k),\ 0 < c < 1.$$

1. Start with a candidate step. If it fails, **backtrack** (shrink it).
2. If it succeeds, **extrapolate** (expand it) to approach the acceptance boundary.
3. Set $w_{k+1} = w_k - \eta_k g_k$ and repeat, updating the reference R_k.

Since R_k can exceed today's loss, an accepted step can temporarily increase f. **Worked example: accepting a temporary loss increase**

Let $f(x) = \frac{1}{2}x^2$, $x = 1$, $g = 1$, and propose $\eta = 2.1$. Then

$$x^+ = 1 - 2.1 = -1.1, \qquad f(x^+) = 0.605 > f(1) = 0.5.$$

A *monotone* rule would reject the step. But with past-loss reference $R = 0.70$ and $c = 0.01$, the nonmonotone right-hand side is $0.70 - 0.01(2.1) = 0.679$. Since $0.605 \le 0.679$, it is accepted. This illustrates the acceptance mechanism, *not* a guarantee that repeated $\eta = 2.1$ is safe.

> **Theorem 2.9 (what is proved).** If $f \in C^1$ is bounded below, its initial level set is compact, and the specified Zhang–Hager-type nonmonotone equality search is used, then $\|\nabla f(w_k)\| \to 0$. This says **first-order stationarity**; it does not assert convergence to a global optimum or avoidance of all saddles.

Additional theoretical properties

Proposition 2.5 gives a *local smoothness-aware lower bound* $\eta_k \ge 2\delta(1-c)/\ell(w_k, \eta_{w_k})$. Proposition 2.8 extends adaptivity to Hölder segment-smooth losses. Corollary 2.10 gives an asymptotic bound involving Hessian eigenvalue *magnitudes*, assuming iterates converge to a stationary point.

3. The surprising finding: the flattest region may be wrong

In the authors' K-class **mean-squared-error** (MSE) classification setting, consider a network output $z = Wh + b$ with an unconstrained final-layer bias $b \in \mathbb{R}^K$, and loss normalized as the mean squared error across K outputs.

A short derivation of the $2/K$ sharpness bound

For a single example, $f(b) = K^{-1} \sum_{j=1}^{K} (z_j - y_j)^2$. Differentiating once in the last-layer bias,

$$\frac{\partial f}{\partial b_j} = \frac{2}{K}(z_j - y_j), \qquad \frac{\partial^2 f}{\partial b_j \partial b_l} = \frac{2}{K}\mathbf{1}_{j=l}.$$

Thus the bias block of the full Hessian is

$$H_{bb} = \frac{2}{K} I_K \quad \Longrightarrow \quad \lambda_{\max}(H_{\text{full}}) \geq \frac{2}{K}.$$

The inequality follows by restricting the Rayleigh quotient to directions changing only b. It is a *global lower bound on sharpness*, **not a lower bound on training loss**. For $K = 10$, the threshold is 0.2; for $K = 100$, it is 0.02.

> **Crucial distinction.** A point minimizing $\lambda_{\max}(H)$ globally is not necessarily a minimum of f. The experiments find points whose sharpness reaches $2/K$ yet whose Hessian also has negative eigenvalues: nearly stationary **saddles** where useful feature learning has stopped.

How the experiments reveal the problem

The authors run full-batch GD on 5,000-image subsets of CIFAR-10, CIFAR-100, SVHN and EMNIST, with MLP, CNN, ResNet, VGG, DenseNet and other architectures across main and supplementary experiments. The main results use MSE; other settings appear in appendices.

Method / phenomenon	What the paper reports
NLS / PoNLS	Large adaptive steps, EoS-like loss oscillations; some runs approach sharpness $2/K$ very early.
Unsuccessfully flat	Sharpness hits $2/K$ early, but loss remains near random-guess performance; hidden-layer gradients are tiny.
Successfully flat	Slightly greater sharpness, much better training loss and often higher test accuracy.
NLS-ub	Caps the step size below K for this K-class MSE setup; empirically avoids the pathological globally-flat region in tested runs.

Why cap $\eta_k < K$?

A self-stabilization *heuristic* suggests limiting sharpness near $2/\eta_k$. If $\eta_k \approx K$, this points toward the smallest possible sharpness $2/K$; capping $\eta_k < K$ leaves room for a slightly sharper valley. This is a setting-specific intervention motivated by the experiments, *not* a universal saddle-escape theorem.

4. Contributions, boundaries, and survey synthesis

What this paper contributes

1. A definition of *large* GD steps based on local segment smoothness (including Hölder generalizations), not global L-smoothness.
2. A practical first-order equality line search combining backtracking and extrapolation, with a convergence-to-stationarity result under explicit assumptions.
3. Empirical evidence that such adaptive steps can operate in EoS-like regimes from early training, and can uncover globally least-sharp but unsuccessful saddle regions.
4. A $2/K$ lower bound on Hessian sharpness in the stated MSE/final-bias setting, and an experimentally motivated step-size cap to avoid premature flatness.

Important assumptions and limitations

- **Theorem vs. practice:** The main convergence theorem gives $\|\nabla f(w_k)\| \to 0$ with C^1, lower-bounded objective, compact starting level set, and the specified nonmonotone reference; no global-minimum or second-order guarantee.
- **Local smoothness:** Stronger step-size bounds need segment/Hölder smoothness and algorithm parameters; experimental choices $c = 10^{-4}$ and $\delta = 0.5$ do not automatically satisfy every theoretical "large" condition.
- **Geometry:** The $2/K$ sharpness bound depends on the MSE normalization and an unconstrained linear final layer with bias. It is not a universal numerical sharpness threshold.
- **Evidence:** Major experiments use 5,000 training images; the authors explicitly note that larger-scale comparisons are needed. Sharpness is a proxy, and compute/runtime dominance is not proved.
- **SGD:** Additional SGD experiments are presented, but a corresponding general convergence theorem is left open.

Running cross-paper comparison: what each answers

Paper / key	Tags	Link to Flatland
Saxe (2014) `saxe2014exact`	D, A / F, B	Initialization and depth can produce long plateaus without bad minima.
Dauphin (2014) `dauphin2014identifying`	G, E / F, B	Saddle curvature causes slow training; curvature-aware intervention.
Lee (2016) `lee2016gradient`	G, D / F	Avoids convergence to strict saddles under conditions, not necessarily quick passage near them.
Jin (2017) `jin2017escape`	D, E / F, B	Perturbations provide finite-time second-order stationarity; Flatland targets step-size dynamics.
Li (2018) `li2018visualizing`	G, A / F, B	Loss flatness is difficult to compare and interpret visually; Flatland adds failure cases.
Du (2019) `du2019gradient`	D, A / F, B	Overparameterization gives a specialized global training-loss convergence result.
Galli (2026) `galli2026flatland`	**D, G, E / R, B**	Large steps can yield EoS and problematic globally flat saddles; cap steps in tested setting.

Quick reference and Papers Master note

GD: $w^+ = w - \eta \nabla f$ **EoS diagnostic:** $\eta \lambda_{\max}(H) \approx 2$ **Line search:** $f(w^+) \le R - c\eta\|\nabla f\|^2$

Sharpness lower bound (MSE): $\lambda_{\max}(H) \ge 2/K$ **Guarantee:** $\|\nabla f(w_k)\| \to 0$.

Our notes (copy to sheet). Galli et al. (2026) define locally smoothness-aware large GD steps and construct nonmonotone equality line searches that converge to first-order stationarity under compact-level-set assumptions. In MSE neural-network experiments, aggressive steps can reach EoS and globally least-sharp ($2/K$) saddle regions prematurely, stalling training; capping steps below K improves tested runs. No global-minimum or general SGD guarantee.

Original: PMLR 306 (ICML 2026). Full text used: arXiv:2606.06722. Numerical scalar examples are pedagogical adaptations.